Explicit Local Langlands for Wildly Ramified Groups: A CRMLint Squeeze

(Paper 78, Constructive Reverse Mathematics Series)

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Abstract

We apply the CRMLint Squeeze protocol (Paper 77) to the Local Langlands Correspondence (LLC) for wildly ramified supercuspidal representations of $\mathrm{GL}(n, F)$ over p-adic fields F.

The Classical Boundary Node is the Harris–Taylor existence theorem [1]: for every irreducible smooth representation π, a Weil–Deligne parameter $\sigma(\pi) : W_F \to \mathrm{GL}(n, \mathbb{C})$ exists. This is CLASS: the proof uses Shimura varieties, étale cohomology, and Lefschetz fixed-point theorems.

The constructive target: for supercuspidal representations built from Bushnell–Kutzko simple types (λ, J) [2], the Galois parameter is determined by finitely many Harish-Chandra character values. These values are computable from the type data by Bushnell–Henniart's explicit matching [3]. The Squeeze reduces the LLC to a polynomial identity over a finite extension of $\mathbb{Q}_p$—deterministically solvable by exact p-adic algebra via asymmetric offloading (Python CAS computes; Lean verifies).

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*Lean 4 formalization available at (DOIpending).

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1 Introduction

1.1 The Local Langlands Correspondence

The LLC for $\mathrm{GL}(n, F)$ establishes a bijection between isomorphism classes of irreducible smooth representations of $\mathrm{GL}(n, F)$ and n-dimensional Frobenius-semisimple Weil–Deligne parameters of W_F. For $n = 1$ (class field theory) and tamely ramified representations ($p \nmid n$), constructive proofs exist. The wildly ramified case ($p \mid n$) requires the full Harris–Taylor machinery.

1.2 The CRMLint Squeeze applied to LLC

Paper 75 [5] proved that the continuous topological machinery (Shimura varieties, v-sites, injective envelopes) in the Fargues–Scholze proof is strictly CLASS. This guarantees that the discrete algebraic framework of Bushnell–Kutzko types provides a BISH path to the same classification—if the character matching can be made explicit.

The Squeeze strategy:

1. **Scaffold:** Formalise the type-theoretic side of LLC in Lean 4 (simple strata, simple types, compact induction).
2. **X-Ray:** CRMLint identifies the CBN as Harris–Taylor's global-to-local transfer.
3. **Excise:** Remove the CBN. The remaining goal is: given (λ, J), construct σ such that $\mathrm{tr}(\sigma(\mathrm{Frob}^k)) = \Theta_\pi(\gamma_k)$ for all test elements γ_k.
4. **Squeeze:** Compute both sides explicitly from the type data. Verify the polynomial identity by `native_decide`.

1.3 Main results

This paper is a scaffold. The explicit computation will follow the asymmetric offloading protocol established in Paper 77.

1.4 Position in the atlas

Paper 78 applies the CRMLint Squeeze to the Langlands program, the deepest interface between number theory and representation theory. If successful, it provides the first formally verified constructive LLC for a wildly ramified supercuspidal representation.

2 Preliminaries

2.1 The CRM hierarchy

$$\mathsf{BISH} \subset \mathsf{BISH{+}MP} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}.$$

2.2 p-adic fields

Let $F/\mathbb{Q}_p$ be a finite extension with ring of integers $\mathcal{O}_F$, maximal ideal $\mathfrak{p}$, residue field $k_F = \mathcal{O}_F/\mathfrak{p} \cong \mathbb{F}_q$ where $q = p^f$, and uniformizer ϖ.

2.3 Bushnell–Kutzko types

Definition 2.1 (Simple stratum)**.** A *simple stratum* $[\mathfrak{A}, n, r, \beta]$ consists of:

- a hereditary order $\mathfrak{A}$ in $M_n(F)$,
- integers $n > r \geq 0$ (the level and jump),
- an element $\beta \in M_n(F)$ with $v_{\mathfrak{A}}(\beta) = -n$, generating a field extension $E = F[\beta]/F$ of degree n.

Definition 2.2 (Simple type)**.** A *simple type* (λ, J) is an irreducible representation λ of a compact open subgroup $J \subset \mathrm{GL}(n, F)$, constructed from a simple stratum by a sequence of Heisenberg extensions and β-extensions.

The key theorem of Bushnell–Kutzko [2]: every supercuspidal representation π of $\mathrm{GL}(n, F)$ arises as $\pi = \text{c-Ind}_J^G(\lambda)$ for some simple type (λ, J). This construction is entirely BISH: it involves only finite group theory and linear algebra over finite extensions of F.

2.4 The Harris–Taylor CBN

Axiom 2.3 (Harris–Taylor 2001)**.** For every irreducible smooth representation π of $\mathrm{GL}(n, F)$, there exists a Frobenius-semisimple Weil–Deligne parameter $\sigma(\pi) : W_F \to \mathrm{GL}(n, \mathbb{C})$ such that:

$$L(s, \pi) = L(s, \sigma(\pi)), \qquad \varepsilon(s, \pi, \psi) = \varepsilon(s, \sigma(\pi), \psi).$$

CRM classification: CLASS. The proof uses the cohomology of Shimura varieties (global input), Lefschetz fixed-point (topological), and trace formula arguments (analytic continuation over $\mathbb{C}$).

3 The Squeeze Architecture

3.1 The character polynomial

For a supercuspidal $\pi = \text{c-Ind}_J^G(\lambda)$, the Harish-Chandra character Θ_π is a locally constant function on the regular elliptic set of $\mathrm{GL}(n, F)$. Bushnell–Henniart [3] gave an explicit formula for $\Theta_\pi(\gamma)$ in terms of the type data (λ, J).

3.2 The Galois trace

On the Galois side, the parameter σ is determined by its characteristic polynomial:

$$\det(1 - \sigma(\mathrm{Frob}) \cdot X) = \prod_{i=1}^{n} (1 - \alpha_i X),$$

where the $\alpha_i \in \overline{\mathbb{Q}}_\ell$ are the Satake parameters. For supercuspidals, these are determined by the simple stratum.

3.3 The matching

The Squeeze reduces to verifying the identity:

$$\sum_{k=1}^{n} \alpha_i^k = \Theta_\pi(\gamma_k) \qquad \text{for } k = 1, \ldots, n,$$

where γ_k ranges over a standard set of test elements. Both sides are explicitly computable from the type data.

4 CRM Audit

Component	CRM Level	Justification
Simple strata, types	BISH	Finite algebra over F
Harris–Taylor existence	CLASS	Shimura varieties
Harish-Chandra character	BISH	Explicit formula
Character polynomial match	BISH	Exact p-adic algebra

4.1 What descends, from where, to where

CLASS (Harris–Taylor for $\mathrm{GL}(n, F)$) $\longrightarrow$ BISH (explicit character polynomial identity).

5 Discussion

Paper 78 extends the CRMLint Squeeze from algebraic geometry (Paper 77: Hodge theory on E^4) to the Langlands program. The key structural parallel: in both cases, a CLASS existence theorem (Lefschetz $(1,1)$ / Harris–Taylor) is replaced by an explicit BISH computation (cup product decomposition / character polynomial matching).

The wildly ramified case ($p \mid n$) is the frontier: for tame representations, the LLC is constructive by Henniart's explicit formulas. The wild case requires navigating Bushnell–Kutzko's β-extensions, which are algebraically intricate but strictly BISH.

6 Conclusion

This paper is a scaffold for the CRMLint Squeeze applied to the Local Langlands Correspondence. The architecture is in place; the explicit computation awaits.

Acknowledgments

Lean 4 formalization produced using AI code generation (Claude Code) under human direction. The mathematical foundations are due to Bushnell–Kutzko [2], Harris–Taylor [1], and Bushnell–Henniart [3].

This work is part of the Constructive Reverse Mathematics series (Papers 1–78). The Lean 4 code uses the Mathlib library; see [9]. The series format follows the CRM format guide [8].

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